REAL NUMBERS: 30 Questions: 40 Minutes

Q1. Which statement is TRUE?
A. Every irrational number is a non-terminating repeating decimal
B. Every terminating decimal is rational
C. Every real number is irrational
D. Every integer is irrational

Q2. Which of the following is NOT a rational number?
A. 0.75
B. √9
C. √2
D. 3/4

Q3. Which statement about decimal fractions is correct?
A. All infinite decimals are irrational
B. All repeating decimals are irrational
C. All terminating decimals are rational
D. All rational numbers are terminating decimals

Q4. Which of the following must be irrational?
A. √16
B. √(25/36)
C. √3
D. √0.81

Q5. Which of the following is a finite decimal?
A. 1/3
B. 1/7
C. 1/4
D. 2/9

Q6. Which statement is FALSE?
A. Rational numbers can be expressed as p/q
B. Irrational numbers cannot be written as p/q
C. Every real number is either rational or irrational
D. Every irrational number is a repeating decimal

Q7. Which of the following is an example of infinite non-repeating decimal?
A. 0.3333...
B. 0.272727...
C. 0.1010010001...
D. 0.25

Q8. Which statement is TRUE about integers?
A. All integers are natural numbers
B. All integers are rational numbers
C. Integers include fractions
D. Integers are always positive

Q9. Which of the following is a non-terminating repeating decimal?
A. 0.125
B. 0.333...
C. √2
D. 5

Q10. Which statement is correct regarding real numbers?
A. Real numbers include only rational numbers
B. Real numbers include only irrational numbers
C. Real numbers include both rational and irrational numbers
D. Real numbers exclude integers

Q11. Which of the following is always rational?
A. Sum of two irrational numbers
B. Product of two irrational numbers
C. Sum of two rational numbers
D. Product of rational and irrational

Q12. Which of the following numbers is both rational and non-terminating?
A. √2
B. 1/3
C. 0.5
D. π

Q13. Which statement is TRUE?
A. All repeating decimals are irrational
B. All repeating decimals are rational
C. All irrational numbers are repeating
D. All integers are irrational

Q14. Which of the following is NOT a real number?
A. √2
B. -5
C. 0
D. None of the above

Q15. Which statement is correct?
A. Irrational numbers can be expressed as ratio of integers
B. Rational numbers include terminating and repeating decimals
C. All infinite decimals are rational
D. All integers are irrational

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Q16. Which of the following best explains why √2 is irrational?
A. It is not a perfect square
B. It cannot be expressed as a terminating decimal
C. It cannot be written as a ratio of two integers
D. It has infinite digits

Q17. If a decimal expansion is non-terminating but repeating, then the number must be:
A. Irrational
B. Rational
C. Integer
D. Prime

Q18. Which of the following numbers is an infinite non-repeating decimal?
A. 0.252525...
B. 0.121212...
C. 0.123456789101112...
D. 0.75

Q19. Which of the following can be expressed as a terminating decimal?
A. 7/12
B. 5/8
C. 2/11
D. 13/99

Q20. Which of the following fractions will produce a repeating decimal?
A. 1/2
B. 1/5
C. 1/6
D. 1/8

Q21. Which statement is TRUE about decimal fractions?
A. All infinite decimals are irrational
B. All repeating decimals are rational
C. All rational numbers are terminating decimals
D. All irrational numbers are finite

Q22. If a number is written as 0.333..., then its equivalent fraction is:
A. 1/2
B. 1/3
C. 2/3
D. 3/10

Q23. Which of the following numbers is rational but not terminating?
A. √2
B. π
C. 2/7
D. √5

Q24. Which statement is FALSE?
A. Every integer is a rational number
B. Every rational number is a real number
C. Every irrational number is a real number
D. Every real number is rational

Q25. Which of the following is a mixed repeating decimal?
A. 0.7777...
B. 0.121212...
C. 0.123333...
D. 0.5

Q26. If a fraction has denominator of the form 2^m × 5^n, then its decimal form will be:
A. Infinite repeating
B. Infinite non-repeating
C. Terminating
D. Irrational

Q27. Which of the following statements about real numbers is correct?
A. They include only integers
B. They include only fractions
C. They include both rational and irrational numbers
D. They exclude decimals

Q28. Which of the following numbers lies between √3 and 4?
A. 1
B. 2
C. 3
D. 5

Q29. Which of the following is an irrational number expressed in decimal form?
A. 0.5
B. 0.333...
C. 0.4142135...
D. 0.25

Q30. Which statement is TRUE regarding infinite decimals?
A. All infinite decimals are irrational
B. Infinite repeating decimals are rational
C. Infinite decimals are always integers
D. Infinite decimals cannot be compared

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SQUARE ROOT, HCF, LCM, DIVISIBILITY: 30 Questions: 40 Minutes

Q1. The number of digits in √47089 is:
A. 2
B. 3
C. 4
D. 5

Q2. Which of the following is a perfect square?
A. 2345
B. 33333
C. 1089
D. 1026

Q3. If a number ends with digit 8, it:
A. Can be a perfect square
B. Cannot be a perfect square
C. Must be prime
D. Must be composite

Q4. √0.002916 = ?
A. 0.054
B. 0.54
C. 0.0054
D. 5.4

Q5. If √x = 48, then x = ?
A. 2304
B. 2408
C. 2500
D. 2116

Q6. √(64/81) = ?
A. 8/9
B. 9/8
C. 16/27
D. 4/9

Q7. Which fraction is a perfect square?
A. 50/32
B. 25/16
C. 18/49
D. 12/25

Q8. Which is prime?
A. 21
B. 35
C. 29
D. 51

Q9. Which is composite?
A. 13
B. 17
C. 19
D. 21

Q10. Which pair is co-prime?
A. (8,16)
B. (9,27)
C. (10,21)
D. (12,18)

Q11. If two numbers are co-prime, their HCF is:
A. 0
B. 1
C. Prime
D. Equal

Q12. Which number is divisible by 3?
A. 148
B. 147
C. 152
D. 146

Q13. Which number is divisible by 9?
A. 378
B. 372
C. 374
D. 376

Q14. A number divisible by both 2 and 3 must be divisible by:
A. 4
B. 5
C. 6
D. 9

Q15. Which is divisible by 4?
A. 3512
B. 3514
C. 3516
D. 3518

Q16. Which number is divisible by 8?
A. 1234
B. 1240
C. 1252
D. 1266

Q17. The greatest number that divides 57, 98, and 183 exactly is:
A. 1
B. 3
C. 7
D. 19

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Q18. The greatest number that divides 365 and 463 leaving remainder 5 and 7 is:
A. 24
B. 12
C. 6
D. 18
Q19. The smallest number divisible by 16, 24, and 32 is:
A. 96
B. 48
C. 64
D. 128
Q20. The least number which when added to 4639 makes it a perfect square:
A. 6
B. 9
C. 7
D. 8
Q21. A number leaves remainder 5 when divided by 16, 24, and 32. The smallest such number is:
A. 91
B. 101
C. 85
D. 96
Q22. If HCF = 10 and LCM = 30, product of numbers =
A. 100
B. 300
C. 200
D. 150
Q23. If product = 3380 and HCF = 13, LCM = ?
A. 260
B. 2600
C. 130
D. 338
Q24. Which cannot be a perfect square?
A. Ends with 6
B. Ends with 4
C. Ends with 7
D. Ends with 1
Q25. If a number has odd number of zeros at the end, it is:
A. Perfect square
B. Not perfect square
C. Prime
D. Co-prime
Q26. Number of digits in √1234321 is:
A. 3
B. 4
C. 5
D. 6
Q27. Smallest number which when multiplied by 900 becomes a perfect square:
A. 2
B. 3
C. 5
D. 6
Q28. Which is NOT co-prime?
A. (8,15)
B. (10,21)
C. (12,25)
D. (9,27)
Q29. A number divisible by 2, 3, and 5 must be divisible by:
A. 10
B. 15
C. 30
D. 60
Q30. Which is true?
A. All primes are co-prime with each other
B. All co-primes are prime
C. Two composite numbers cannot be co-prime
D. HCF of co-prime numbers is 1

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UNITARY METHOD: 20 Questions: 30 Minutes

Q1. A group of 8 people has food for 15 days. After 5 days, 4 more people join. For how many more days will the remaining food last?
A. 5
B. 6
C. 7
D. 8

Q2. 12 workers can complete a work in 18 days. After working 6 days, 6 more workers join. In how many total days will the work be finished?
A. 10
B. 12
C. 14
D. 15

Q3. A man walks at 5 km/h and reaches office 10 minutes late. If he walks at 6 km/h, he reaches 5 minutes early. Find distance.
A. 5 km
B. 6 km
C. 7.5 km
D. 8 km

Q4. A train crosses a pole in 12 sec and a platform (180 m) in 18 sec. Find speed of train.
A. 54 km/h
B. 60 km/h
C. 72 km/h
D. 90 km/h

Q5. A boat goes 30 km downstream in 2 hours and returns upstream in 3 hours. Find speed of stream.
A. 2.5
B. 3
C. 4
D. 5

Q6. Two pipes can fill a tank in 6 hours. One pipe alone takes 10 hours. After opening both, one pipe is closed after 2 hours. Total time to fill tank?
A. 7
B. 8
C. 9
D. 10

Q7. A and B can do a work in 12 days, B and C in 15 days, C and A in 20 days. How long will A, B, and C together take?
A. 8
B. 10
C. 12
D. 15

Q8. A man covers half distance at 6 km/h and remaining at 12 km/h. Find average speed.
A. 8
B. 9
C. 10
D. 12

Q9. A train leaves Dhaka at 60 km/h. Another train leaves 1 hour later at 80 km/h. When will it catch the first train?
A. 2 hr
B. 3 hr
C. 4 hr
D. 5 hr

Q10. A pipe fills tank in 8 hours, another empties in 12 hours. Both open together, but after 2 hours empty pipe is closed. Total time?
A. 10
B. 12
C. 14
D. 16

Q11. 6 men can do a work in 10 days. After 3 days, 3 more men join. In how many total days will work finish?
A. 6
B. 7
C. 8
D. 9

Q12. A boat goes 24 km upstream in 4 hours and same downstream in 2 hours. Find speed of boat in still water.
A. 6
B. 9
C. 10
D. 12

Q13. A train crosses a bridge in 30 sec and a pole in 18 sec. Bridge length = 240 m. Find train length.
A. 120
B. 150
C. 180
D. 200

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Q14. A sum of money feeds 10 people for 20 days. After 5 days, 5 people leave. For how many more days will the food last?
A. 15
B. 20
C. 25
D. 30

Q15. A and B together can do a work in 8 days. A alone takes 12 days. If A works for 3 days, then B finishes, total time?
A. 6
B. 7
C. 8
D. 9

Q16. A person walks first 1/3 distance at 4 km/h, next 1/3 at 6 km/h, last 1/3 at 12 km/h. Find average speed.
A. 6
B. 6.5
C. 7
D. 8

Q17. Two trains 120 m and 180 m long cross each other in 10 sec in opposite directions. Find relative speed.
A. 20 m/s
B. 25 m/s
C. 30 m/s
D. 35 m/s

Q18. A cistern has two pipes. One fills in 6 hr, another in 8 hr. A leak empties tank in 24 hr. How long to fill?
A. 3
B. 4
C. 5
D. 6

Q19. A group of 5 workers can complete a task in 20 days. After 10 days, 5 more workers join. Total time?
A. 12
B. 15
C. 18
D. 20

Q20. A man walks to a place at 4 km/h and returns at 6 km/h. Find average speed.
A. 4.8
B. 5
C. 5.2
D. 5.5

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RATIO, PERCENTAGE, PROFIT-LOSS: 30 Questions: 40 Minutes

Q1. The ratio of two numbers is 3 : 5 and their sum is 64. Find the larger number.
A. 30
B. 35
C. 40
D. 45

Q2. If a : b = 4 : 7 and b : c = 14 : 15, find a : b : c.
A. 8 : 14 : 15
B. 4 : 14 : 15
C. 8 : 7 : 15
D. 4 : 7 : 15

Q3. Divide Tk. 720 in the ratio 2 : 3 : 4. The smallest share is:
A. 120
B. 160
C. 180
D. 200

Q4. The ratio of boys to girls is 5 : 4. If 12 more girls join, the ratio becomes 5 : 6. Find original number of boys.
A. 40
B. 50
C. 60
D. 70

Q5. If a number is increased by 20% and then decreased by 20%, the net change is:
A. No change
B. 4% increase
C. 4% decrease
D. 2% decrease

Q6. Ratio of A to B is 2 : 3. If A increases by 50% and B decreases by 25%, new ratio is:
A. 1 : 1
B. 4 : 3
C. 2 : 1
D. 3 : 2

Q7. Convert the ratio 3 : 4 into percentage.
A. 60%
B. 70%
C. 75%
D. 80%

Q8. If 40% of a number is 120, find the number.
A. 200
B. 240
C. 300
D. 360

Q9. If the ratio of two numbers is 7 : 9 and their difference is 20, find the smaller number.
A. 70
B. 80
C. 60
D. 90

Q10. A price increases from Tk. 500 to Tk. 650. Percentage increase is:
A. 20%
B. 25%
C. 30%
D. 35%

Q11. A shopkeeper gains 20% after giving 10% discount. Find marked price : cost price.
A. 4 : 3
B. 3 : 2
C. 5 : 3
D. 6 : 5

Q12. A trader sells at 10% loss. If SP = Tk. 180, find CP.
A. 190
B. 200
C. 210
D. 220

Q13. Profit is 25%. What is SP : CP?
A. 5 : 4
B. 4 : 5
C. 6 : 5
D. 3 : 2

Q14. A mixture contains milk and water in ratio 5 : 3. If 16 liters water is added, ratio becomes 5 : 7. Find initial milk.
A. 20
B. 25
C. 30
D. 40

Q15. If a : b = 3 : 4, then (a + b) : (a − b) = ?
A. 7 : 1
B. 1 : 7
C. 7 : −1
D. 1 : 3

Q16. If 25% of A = 40% of B, find A : B.
A. 8 : 5
B. 5 : 8
C. 4 : 3
D. 3 : 4

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Q17. The ratio of ages of father and son is 5 : 2. After 10 years, ratio becomes 3 : 2. Find present age of father.
A. 40
B. 50
C. 60
D. 70

Q18. A number is increased by 25% and then decreased by 20%. Net effect:
A. 0%
B. 5% increase
C. 5% decrease
D. 2% increase

Q19. If cost price increases by 20% and selling price remains same, profit decreases by:
A. 20%
B. 25%
C. 16⅔%
D. 10%

Q20. A man spends 60% of his income. If his income increases by 25% and expenditure increases by 20%, savings increase by:
A. 30%
B. 40%
C. 50%
D. 60%

Q21. Ratio of two numbers is 4 : 5. If 8 is added to both, ratio becomes 6 : 7. Find smaller number.
A. 16
B. 24
C. 32
D. 40

Q22. If a : b : c = 2 : 3 : 4, find (a+b) : (b+c).
A. 5 : 7
B. 3 : 4
C. 4 : 5
D. 2 : 3

Q23. Convert 0.75 into ratio form.
A. 3 : 4
B. 4 : 3
C. 2 : 3
D. 5 : 4

Q24. If 30% of a number is equal to 45% of another, find their ratio.
A. 3 : 2
B. 2 : 3
C. 4 : 3
D. 3 : 4

Q25. A man gains 25% by selling at Tk. 500. If he sells at Tk. 440, loss % is:
A. 10%
B. 12%
C. 15%
D. 20%

Q26. The ratio of two numbers is 3 : 4. If each is increased by 10, ratio becomes 5 : 6. Find numbers.
A. 30, 40
B. 15, 20
C. 9, 12
D. 6, 8

Q27. If CP = Tk. 400 and profit = 20%, find SP.
A. 440
B. 460
C. 480
D. 500

Q28. If SP decreases by 10% and CP remains same, profit decreases by:
A. 10%
B. 20%
C. 25%
D. Depends

Q29. A number is first increased by 10% then decreased by 10%. Net effect is:
A. 1% decrease
B. No change
C. 1% increase
D. 2% decrease

Q30. Ratio of shaded to total is 3 : 5. What percentage is shaded?
A. 40%
B. 50%
C. 60%
D. 75%

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SIMPLE & COMPOUND INTEREST: 30 Questions: 40 Minutes

Q1. A sum of Tk. 5,000 gives a profit of Tk. 3,000 in 6 years. Find the rate of profit.
A. 8%
B. 10%
C. 12%
D. 15%

Q2. A sum becomes Tk. 8,000 in 6 years at 10% simple profit. Find principal.
A. 4,000
B. 5,000
C. 6,000
D. 7,000

Q3. Profit of Tk. 2,550 is obtained in 6 years at 8½% simple profit. Find principal.
A. 4,000
B. 5,000
C. 6,000
D. 7,000

Q4. At what rate will Tk. 3,000 become Tk. 4,500 in 5 years (simple)?
A. 8%
B. 10%
C. 12%
D. 15%

Q5. A sum becomes double in 8 years at simple profit. Find rate.
A. 10%
B. 12.5%
C. 15%
D. 20%

Q6. A sum becomes Tk. 5,500 in 3 years. Profit is 3/8 of principal. Find principal.
A. 3,000
B. 4,000
C. 5,000
D. 6,000

Q7. Profit of Tk. 4,800 is earned on Tk. 10,000 at 12% rate. Find time.
A. 2
B. 3
C. 4
D. 5

Q8. Two sums at same rate produce profit Tk. 3,000 and Tk. 1,500 in 5 years. Find ratio of principals.
A. 1 : 2
B. 2 : 1
C. 3 : 1
D. 1 : 3

Q9. A sum becomes Tk. 10,200 in 4 years and Tk. 8,840 in 3 years. Find rate.
A. 8%
B. 10%
C. 12%
D. 15%

Q10. A principal becomes Tk. 28,000 in 3 years and Tk. 30,000 in 5 years. Find principal.
A. 25,000
B. 24,000
C. 23,000
D. 22,000

Q11. A sum is divided into two parts x and y at 6% and 4%. Total interest = Tk. 2,840. Total sum = Tk. 56,000. Find x.
A. 30,000
B. 28,000
C. 26,000
D. 24,000

Q12. A sum becomes double in 6 years. In how many years will it become triple?
A. 10
B. 12
C. 15
D. 18

Q13. Profit of Tk. 1,500 is obtained in 5 years. Principal = Tk. 3,000. Find rate.
A. 8%
B. 10%
C. 12%
D. 15%

Q14. A principal becomes Tk. 6,500 in 1 year and Tk. 6,760 in 2 years. Find principal.
A. 6,000
B. 6,200
C. 6,300
D. 6,400

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Q15. A sum at 10% gives Tk. 200 profit in 1 year. What will be amount in 5 years?
A. 2,800
B. 3,000
C. 3,200
D. 3,500

Q16. Tk. 2,000 gives compound profit at 10%. Find amount after 2 years.
A. 2,400
B. 2,420
C. 2,440
D. 2,450

Q17. Find compound amount of Tk. 8,000 in 3 years at 10%.
A. 10,200
B. 10,648
C. 10,800
D. 11,000

Q18. Find difference between simple and compound profit on Tk. 5,000 in 3 years at 10%.
A. 100
B. 150
C. 155
D. 165

Q19. A principal becomes Tk. 19,500 in 1 year and Tk. 20,280 in 2 years (compound). Find rate.
A. 4%
B. 5%
C. 6%
D. 8%

Q20. A sum becomes Tk. 6,174 after 2 years at compound rate 5%. Find principal.
A. 5,000
B. 5,200
C. 5,600
D. 6,000

Q21. Compound amount after 3 years is Tk. 78,732 at 8%. Find principal.
A. 60,000
B. 62,500
C. 65,000
D. 70,000

Q22. A principal becomes Tk. 2,24,720 in 1 year with half-yearly compounding. Initial was Tk. 2,00,000. Find rate.
A. 10%
B. 12%
C. 14%
D. 16%

Q23. Profit in first year = Tk. 120 on Tk. 1,000 at 12%. Find amount after 3 years (compound).
A. 1,404
B. 1,405
C. 1,410
D. 1,420

Q24. If rate is 12%, what will Tk. 10,000 become in 3 years (compound)?
A. 13,000
B. 14,049
C. 14,405
D. 15,000

Q25. A sum becomes 1.331 times in 3 years. Find rate.
A. 8%
B. 10%
C. 12%
D. 15%

Q26. A sum doubles in 5 years (compound). Find rate.
A. 10%
B. 12%
C. 14%
D. 15%

Q27. Compound profit on Tk. 5,000 at 10.5% in 2 years is:
A. 1,050
B. 1,100
C. 1,105
D. 1,200

Q28. A principal grows to Tk. 1,254 in 3 years at 8%. Find principal.
A. 1,000
B. 1,050
C. 1,100
D. 1,200

Q29. A sum becomes Tk. 61,740 in 2 years at 5% compound. Find principal.
A. 56,000
B. 58,000
C. 60,000
D. 62,000

Q30. If compound amount after 2 years is Tk. 12,100 and rate = 10%, find principal.
A. 10,000
B. 10,500
C. 11,000
D. 11,500

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